Bill of Materials

This bill of materials (BOM) contains **EVERYTHING** needed to build a setup. From the less memorable screw to the super expensive syringes used for reward delivery.

> **NOTE**
> This BOM is destined to the people from the Champalimaud Foundation (CF). Nonetheless, the goal is that people outside CF are also able to order everything and build the setup.

> **WARNING**
> This BOM specifies the part numbers used by the lab for these setups, but not every component needs to be of a specific part number.
>
> The only things that need to be of a specific model are the Harp Devices, the Behavior Poke Port Breakout v1.1, the Poke Small v1.1 and the 12V Power Supplies.
>
> It's possible to use a different camera model, but please confirm that the camera can be used in Bonsai (although for some cameras the Bonsai workflow will probably need to be adapted).

Harp

This section contains the setup components that relate to the Harp devices somehow (except for the Camera that has a section of its own).

Devices

The Harp devices are assembled by the Hardware and Software Platform, so place an Agendo request to order them.

> **WARNING**
> For people outside of the Champalimaud Foundation, it's possible to order the Harp devices in the Open Ephys Production Site.
>
> It's also possible to assemble the devices in-house with the right equipment since all of them are under an open source license. Some devices (like the Harp SyringePump) must be built in-house since they are not sold externally.

Item	Description	Amount	Observations
Harp Behavior	General-purpose Harp board	1	-
Harp SoundCard	Delivers the auditory stimulus	1	-
Harp Audio Amplifiers	Amplifies the auditory stimulus	2	1 Harp Audio Amplifier per speaker
Harp ClockSynchronizer	Synchronizes the timestamps from every Harp device	1	-
Harp SyringePump	Device used for reward delivery	2	-
Harp CurrentDriver	Drives/controls the LED/laser used in optogenetics	1	Optional

Peripherals

For this part of the BOM, a peripheral is considered to be anything that either interacts with the Harp devices or is needed make the different devices work and/or interact with each other.

Item	Description	Amount	Part Number	Part of the Harp Device kit	Agendo	Observations
Behavior Poke Port Breakout v1.1	Makes the ethernet ports pins from the Harp Behavior available	3	-	❌	✅	Connects to the Harp Behavior
RJ-to-RJ cables	-	3	IM1037	❌	❌	Connects the Harp Behavior to the Behavior Poke Port Breakout v1.1
Poke Small v1.1	Board with infrared beam to detect animal pokes	3	-	❌	✅	Connects to the Behavior Poke Port Breakout v1.1

Item	Description	Amount	Part Number	Part of the Harp Device kit	Agendo	Observations
3.5 mm-stereo-audio-jack-to-bare-wires	-	3	*TODO*	❌	❌	Connects the Poke Small v1.1 to the Behavior Poke Port Breakout v1.1
5 mm white/blue LED	Placed on the box lid to give cues to the animal	1	-	❌	❌	Connects to the LED0 pins of the Harp Behavior
3 mm green LED	Placed in the central poke to give cues to the animal	1	-	❌	❌	Connects to the LED1 pins of the Harp Behavior
RCA-to-RCA cables	-	2	-	✅	❌	Connects each Harp Audio Amplifier to the Harp SoundCard
Speakers	Deliver the auditory stimulus	2	*TODO*	❌	❌	1 speaker per Harp Audio Amplifier
Banana Plug	4 mm Triple Contact Plug (Black or Red) - Pack of 10 units	4	557-0100	❌	❌	Connects the speakers to the Harp Audio Amplifiers
10 ml Glass Syringe	Hamilton 1000 Series Gastight Syringes: Luer Lock Syringes,	2	Hamilton 81620	❌	❌	Used for the Harp SyringePump

Item	Description	Amount	Part Number	Part of the Harp Device kit	Agendo	Observations
	TLL Termination					
4-way Stopcock	Pack of 10 units	2	Masterflex 30600-04	❌	❌	Attaches to the end of the syringe (1 per Harp SyringePump)
Nylon Male Luer Fitting	Pack of 25 units	2	Masterflex MFLX45505-31	❌	❌	Attaches to one end of the stopcock (1 per Harp SyringePump)
Nylon Female Luer Fitting	Pack of 25 units	2	Masterflex 45502-00	❌	❌	Attaches to one end of the stopcock (1 per Harp SyringePump)
Spouts	Used in reward delivery	2	B14200 100 BULK	❌	❌	Used in reward delivery. Glued to the physical lateral pokes
Flexible Tubing	Masterflex Tygon E-3603 Non-DEHP Tubing - 15 meters per unit	2	Masterflex 06407-71	❌	❌	Connects the spouts to one of the luer fittings (either male or female)
Mini USB cable	-	5/6*	AK-300130-018-S	✔	❌	Connects the Harp boards to the computer
Micro USB cable	-	1	-	✔	❌	Used to upload sounds to the Harp SoundCard
3.5 mm-stereo-	-	4/5*	*TODO*	❌	❌	Connects the Harp ClockSynchronizer to

Item	Description	Amount	Part Number	Part of the Harp Device kit	Agendo	Observations
audio-jack-to-jack cable						every other Harp device
12V Power Supply**	AC/DC Wall Mount Adapter 12V 12W	9/10*	VER12US120-JA	✔***	✘	1 is used to power the LED strip that illuminates the behavior box, the remaining are used to power the Harp devices

* if the Harp CurrentDriver is being used

** the number of power supplies can be decreased as explained in the subsection below

*** the power supply used to power the LED strip that illuminates the behavior box must be ordered separately

Reducing the number of power supplies

From the table above, it's pretty noticeable that a lot of power supplies are required, which takes up a lot of space and power outlets. It's possible to use the same power supply for different devices according to their characteristics. The devices can be grouped in the following way:

- Harp Behavior, Harp SoundCard and Harp CurrentDriver*
- V+ of the Harp Audio Amplifiers
- V- of the Harp Audio Amplifiers
- Both Harp SyringePumps

From the list above, it's possible to switch the last line from the previous table with the lines from the table below.

Item	Description	Amount	Part Number	Comes with Harp Device	Agendo	Observations
12V Power Supply	AC/DC Wall Mount Adapter 12V 12W	5	VER12US120-JA	✔***	✘	1 is used to power the LED strip that illuminates the behavior box, the remaining are used to power the Harp devices
2-to-1 cables for the 12V power supplies	-	4/3*		✘	✔***	-
4-to-1 cables for the 12V power supplies	-	0/1*		✘	✔***	-

* if the Harp CurrentDriver is being used

Mechanical Components

This section contains the mechanical components of the setup that were developed and assembled in-house.

> **WARNING**
>
> The files for the mechanical components are not currently available online and can't also be ordered, so people outside of the Champalimaud Foundation will have to develop and assemble their own.

Item	Description	Amount	Part Number	Observations
Behavior box	-	1	-	Made of acrylic
Physical pokes	-	3	-	Preferably made of metal, but can also be 3D printed
Speaker holder	-	2	-	Preferably made of metal, but can also be 3D printed
Speaker holder pole	Hollow alluminium tube	2	-	-
Box LED holder	-	1	-	3D printed

Camera

This section contains the hardware needed to setup the camera and fix it to the lid of the behavior box. Click here to go to the camera configuration instructions.

Item	Description	Amount	Part Number	Observations
FLIR Camera		1	BFS-U3-16S2M-CS	-
Camera Lens		1	A4Z2812CS-MPIR	-
Camera USB cable	USB-A to Micro-B Straight with Thumbscrews, 3 Meters	1	CEI USB3-1-1-2-3M	This cable MUST be connected to a USB 3.0 port for performance
Camera GPIO cable	6 Pin Female Straight Plug (Hirose HR10A-7P-6S) to Flying Leads, 3 Meters	1	CEI MVA-50-3-X-3	Connects to the Harp Behavior to trigger/monitor the camera frames
Tripod Adapter	BFS 30 mm BFLY CM3 Tripod Adapter	1	ACC-01-0003	Used to fix the camera to the Thorlabs poles in the behavior box
Thorlabs 75mm post	Ø12.7 mm Optical Post, SS, M4 Setscrew, M6 Tap, L = 75	1	TR75/M	Connects to the camera

Item	Description	Amount	Part Number	Observations
	mm			
Thorlabs 150mm post	Ø12.7 mm Optical Post, SS, M4 Setscrew, M6 Tap, L = 150 mm	1	TR150/M	Fixes the camera structure to the box lid
Thorlabs post clamp	Rotating Clamp for Ø1/2" Posts, 360° Continuously Adjustable, 5 mm Hex	1	SWC/M	Fixes both Thorlabs posts to each other
M6 Screw	15/16 mm	1		Fixes the longer post to the behavior box lid
M6 Setscrew	M6 x 1.0 Stainless Steel Setscrew, 16 mm Long, 25 Pack	1	SS6MS16	Connects the Thorlabs post to the Camera

Others

This section contains the remaining components needed for the setup. None of these components needs to be the exact model present on the list. It just corresponds to the models that have been used by the lab, but feel free to use different ones, as long as they work.

Item	Description	Amount	Part Number	Observations
Computer	Mini PC Blackview MP100 Mini PC AMD Ryzen 7 5825U/16GB/512GB SSD	1	MP100(16+512)-BLACK	-
Screen	-	1	-	-
HDMI cable	-	1	-	-
Keyboard	-	1	-	-
Mouse	-	1	-	-
USB Hub	HUB USB 3.0 TP-Link UH700 7 Ports	1	P004616	-
Power extension with 6 electrical	Must be appropriate to 90° plugs	1		To plug all of the power supplies in a setup

Item	Description	Amount	Part Number	Observations
outlets				
KVM switch		-		Optional: useful in case one wants to use the same screen + mouse + keyboard kit in different computers
12V LED Strip		1		-
Optogenetics Light Source	-	-	-	Optional

Software Installation

The task's software is composed by the Bonsai workflow, which contains the task logic, and by small Python scripts that perform some operations at the beginning and at the end of the a session (namely, parsing both input and output files from/to more human readable formats). In order to facilitate the deployment of the project and with reproducibility in mind, the Bonsai workflow was developed inside a Bonsai environment and the Python scripts uses uv to create and maintain the Python virtual environment.

To install the task's software, follow the steps:

1. Download the source code from the latest project release and unzip it. Alternatively, clone the repository.
2. Run `./Setup.cmd` to install the Bonsai environment and the Python environment. An application window will appear which is used to specify the COM port each Harp device corresponds to and the paths to the configuration files and the output directory.
3. Download the configuration files templates (`animal.yml`, `setup.csv`, `training.csv`) from the release mentioned in step 1 and place them wherever it's more convenient.
4. After configuring every parameter present in the application, click on the `Update Configuration` button. The button will generate the `./src/config/config.yml` file, which is a hard-coded file used by both the Python scripts and the Bonsai workflow. Close the application window.

> **CAUTION**
>
> Please don't move or delete the `./src/config/config.yml` file! If, for some reason, any of the paths or COM ports need to be changed, re-run the `Setup.cmd` script.

Additional Software

In order to use the FLIR camera to record the sessions, the Spinnaker drivers MUST be installed as well. It's mandatory that the computer has the version 1.29.0.5 of the drivers installed, since the version supported by Spinnaker Bonsai Package. After opening the installer, follow the steps below:

1. Click on `Next`. Then, accept the terms and click on `Next` again.
2. Select `Application Development` and click on `Next`.
3. Deselect `GigE Driver` and click on `Next`.
4. Deselect the `I will use GigE Cameras.` checkbox and click on `Next`.
5. Finally, click on `Install`.

Harp Devices Setup

In order to use the Harp devices, the USB drivers and each devices's firmware must be installed. The installer for the USB drivers can be downloaded here. The Firmware section contains the instructions for installing each board's firmware.

> **NOTE**
> The Harp devices are usually delivered with the firmware already installed, but it may be useful to know how to install the firmware (for example: in case a new firmware version that solves a particular bug is released)

Firmware

For most devices, it's possible to install the corresponding firmware in two different ways.

Firmware Download

1. Go to the Harp Tech GitHub organization.
2. Search for the board's repository. The name of the repository follows the format device.[*board_name*] (for example, device.behavior).
   - **Note:** The Harp Audio Amplifier is not considered a device by itself, but a peripheral for the Harp SoundCard, so there isn't a firmware to be installed for this board.
3. Click on `Releases` and search for the latest firmware release, whose name follows the format fw[*firmware_version*]-harp[*harp_core_version*] (for example, fw2.2-harp1.13).
4. Download the latest version of the firmware binary corresponding to the hardware version of the board being used. The firmware binary name follows the format [*board_name*]-fw[*firmware_version*]-harp[*harp_core_version*]-hw[*hardware_version*]-ass[*assembly_version*].hex (for example, Behavior-fw3.2-harp1.13-hw2.0-ass0.hex).
   - **Note:** For the Harp SoundCard, an additional firmware binary must be downloaded (the PIC32 firmware).

Via Harp Convert to CSV GUI

If the Harp Convert to CSV GUI is already installed, skip to step 3. If it's not already installed, but other Labview-based Harp board GUI is, skip to step 2 instead.

1. Install the LabView Runtime and reboot the computer.
2. Install the latest version of the Harp Convert to CSV GUI.
3. Open the Harp Convert to CSV GUI.
4. Click on `Options` and write "bootloader" in the `List` textbox. The *Update Firmware* window should appear.

5. Choose the communication port (*COMx*) of the board whose firmware is going to be installed.
6. Select the firmware binary to be installed and click on `Update`.
    - **Note:** For the Harp SoundCard, during the installation of the firmware, select the PIC32 firmware when the application asks for the 32 bits device firmware.

Via Bonsai

As explained at the beginning of the section, it is not possible to install the Harp SoundCard firmware through this method. Skip to step 4 if the Bonsai environment is already installed.

> **CAUTION**
>
> It is not possible to use this method to install the firmware of the Harp SoundCard.

1. Install Bonsai.
2. Open Bonsai and click on `Manage Packages`.
3. Install the following packages:
    - Bonsai - Harp Library
    - Bonsai - Harp Design Library
4. Start a new workflow and add a Device node from the Harp package to it.
5. Change the `PortName` property of the node to the communication port (*COMx*) of the board whose firmware is going to be installed.
6. Double-click on the Device node. The *Device Setup* window should appear.
7. Click on `Bootloader>>`. The *Device Setup* should expand.
8. Click on `Open...` to select the firmware binary to be installed and then click on `Update`.

Calibration Protocols

Some of the hardware used in the task requires calibration. Since the calibration protocols are already written somewhere else, they can be found by clicking in the name of the protocols from the list below:

- Speaker Calibration
- Harp SyringePump Calibration
- Optogenetics LED Calibration (*Not available yet*)

Camera Configuration

The setups need a camera to record the sessions. At the moment, the cameras being used are FL3-U3-13S2M from Point Grey. These cameras need to be configured so that they meet the following requirements:

- the frame rate is 100 fps.
- the camera sends a digital signal to the Harp Behavior that signals when each frame is acquired.

After installing the program, connect the camera to the computer in an USB-3.0 port (connecting it to an USB-2.0 port limits the camera's capabilities, namely the frame rate).

TODO: configuration itself and strobe cable

Task Configuration

After setting up both the hardware and the software needed for the task, there's just one more step to follow before start running the task: configuring it.

Despite the task having a common structure, there are small variations to the task that can be achieved by tweaking inumerous configurations. Additionally, there's a need to input the calibration parameters for different pieces of hardware (speakers, SyringePumps, etc), which vary from setup to setup. Because of this, 3 different configuration files were created:

- `animal.yml` - This file contains task configurations that are not level-dependent and/or don't change throughout a session. Some of the parameters present in this file can be updated based on the previous session. A description of each animal-specific setting can be found here.
- `training.csv` - Generally, the configurations that can be set in this file are task parameters that can change a lot during the training phase of an animal. Each line of the file is a different training level the animal has to progress to/through. It can be assumed that when the animal reaches the last level, it's ready for the "actual" experiment. A description of each training-specific setting can be found here.
- `setup.csv` - The settings that can be found in this file don't usually change the logic of the state machine, but are necessary for the setup to be working correctly (for example: equipment calibration parameters). Each line of this file is a different setup, so it's possible to have a single file stored in a drive containing the configurations for every setup and "point" to that file from the pre-configuration script (the application mentioned in the Software Installation page). A description of each setup-specific setting can be found here.

Output Directory

It is intended that all data from every animal is saved in a single directory (the output directory). Look at the following example of an output directory.

```
output/
├── Rat001/
│   ├── 241001/
│   │   ├── out_0.json
│   │   ├── out_0.csv
│   │   ├── video_0.avi
│   │   ├── cam_metadata_0.csv
│   │   ├── events/
│   │   │   ├── behavior_0_32.bin
│   │   │   ├── behavior_0_33.bin
│   │   │   ├── soundcard_0_32.bin
│   │   │   ├── soundcard_0_33.bin
│   │   │   ├── left_pump_0_32.bin
```

```
│   │   │   ├── left_pump_0_33.bin
│   │   │   ├── right_pump_0_32.bin
│   │   │   ├── right_pump_0_33.bin
│   │   │   ├── current_driver_0_32.bin
│   │   │   └── current_driver_0_33.bin
│   │   └── config/
│   │       ├── animal_0.yml
│   │       ├── setup_0.csv
│   │       └── training_0.csv
│   └── 241002/
│       ├── out_0.json
│       ├── out_1.json
│       ├── out_0.csv
│       ├── out_1.csv
│       ├── video_0.avi
│       ├── video_1.avi
│       ├── cam_metadata_0.csv
│       ├── cam_metadata_1.csv
│       ├── events/
│       │   ├── behavior_0_32.bin
│       │   ├── behavior_0_33.bin
│       │   ├── behavior_1_32.bin
│       │   ├── behavior_1_33.bin
│       │   ├── soundcard_0_32.bin
│       │   ├── soundcard_0_33.bin
│       │   ├── soundcard_1_32.bin
│       │   ├── soundcard_1_33.bin
│       │   ├── left_pump_0_32.bin
│       │   ├── left_pump_0_33.bin
│       │   ├── left_pump_1_32.bin
│       │   ├── left_pump_1_33.bin
│       │   ├── right_pump_0_32.bin
│       │   ├── right_pump_0_33.bin
│       │   ├── right_pump_1_32.bin
│       │   ├── right_pump_1_33.bin
│       │   ├── current_driver_0_32.bin
│       │   ├── current_driver_0_33.bin
│       │   ├── current_driver_1_32.bin
│       │   └── current_driver_1_33.bin
│       └── config/
│           ├── animal_0.yml
│           ├── animal_1.yml
│           ├── setup_0.csv
│           ├── setup_1.csv
│           ├── training_0.csv
│           └── training_1.csv
```

```
└── Rat002/
    └── 241003/
        ├── out_0.json
        ├── out_0.csv
        ├── video_0.avi
        ├── cam_metadata_0.csv
        ├── events/
        │   ├── behavior_0_32.bin
        │   ├── behavior_0_33.bin
        │   ├── soundcard_0_32.bin
        │   ├── soundcard_0_33.bin
        │   ├── left_pump_0_32.bin
        │   ├── left_pump_0_33.bin
        │   ├── right_pump_0_32.bin
        │   ├── right_pump_0_33.bin
        │   ├── current_driver_0_32.bin
        │   └── current_driver_0_33.bin
        └── config/
            ├── animal_0.yml
            ├── setup_0.csv
            └── training_0.csv
```

Inside the output directory there is a folder for every animal. Inside each animal's folder there is a folder for every session day, whose name is in the `YYMMDD` format, and inside every session day folder there are different files and folders types:

- `out_X.json` - this is the main output file and has the output data from every trial from every session an animal did so far as Bonsai saves it.
- `out_X.csv` - this is the CSV version of the `out_X.json` file, since it is a more human readable format. This file is generated by a Python script that runs after Bonsai closes.
- `video_X.avi` - this is the CSV version of the `out_X.json` file, since it is a more human readable format. This file is generated by a Python script that runs after Bonsai closes.
- `cam_metadata_X.csv` - this is the CSV version of the `out_X.json` file, since it is a more human readable format. This file is generated by a Python script that runs after Bonsai closes.
- `events/` - this directory has every event from every Harp device connected during a session. The files inside this directory have a name formatted as follows: [*device_name*]_X_[*register*].bin. Basically, there's a different file with Harp messages per device register.
- `config/`- this directory contains a copy of the configuration files used in the current session.

CAUTION

Don't store any other files in the output directory that are not generated automatically during a session and don't change the names of the files. Some operations - auto-update of configuration files, the numbers in the file names - depend on the files saved by the task's software and on the way they are saved.

If there's a need to manually save other files related to the experiment, please do it somewhere else.

Running the Task

After having everything else setup, the only thing left is to run the task. To do it, follow the steps:

1. In the task's directory, double-click on the `./Run.cmd` script.
2. Answer the initial prompt. Then, the bonsai workflow will open and the task will start with the GUI that allows the user to monitor the task in real-time and interact with some aspects of it.

Known Issues

This project is still being actively developed and maintained. However, since this project is composed by a lot of small parts, it's normal that some issues are discovered and, by one reason or another, can't be addressed right away. Here's a list of known issues and how to deal with them for now.

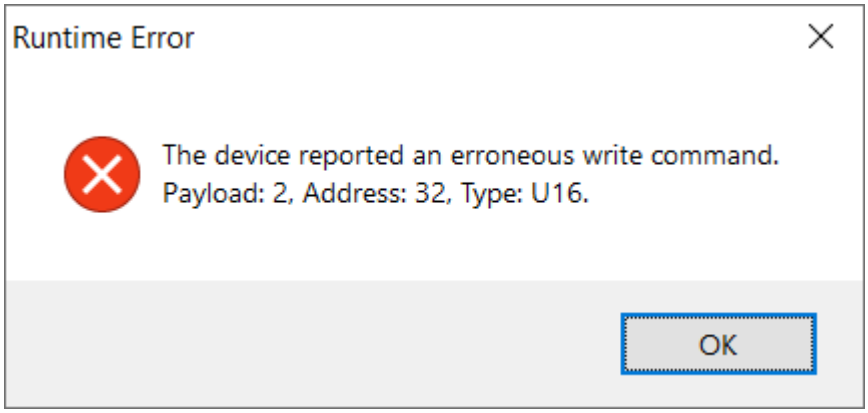
"The Silent Crash"

This issue seems to happen randomly. Basically, what happens is that, for some reason, the workflow stops working at the end of a trial and a new trial never starts, but it's still possible to interact with some of the hardware (for example, it's still possible to deliver water). It's also not possible to "normally" close the workflow, Bonsai must be closed from the Task Manager.

For now, there's no solution to this problem, since a pattern hasn't been found yet. The only recommendation is to close and restart the task.

Harp SoundCard Crash

It's not yet known when or why this crash happens, but it's known that it's an issue related to the Harp SoundCard firmware that makes the whole workflow explicitly crash.



If the crash shown in the above figure occurs, unplug the Harp SoundCard and plug it again before restarting the workflow.